

2008 ENGINE PERFORMANCE

Fuel Charging & Controls - 3.5L - Edge & MKX

SPECIFICATIONS

Material

Item	Specification	Fill Capacity
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-

TORQUE SPECIFICATIONS

Description	Nm	lb-in
Fuel rail bolts	10	89
Throttle Body (TB) bolts	10	89

DESCRIPTION AND OPERATION

FUEL CHARGING AND CONTROLS

Sequential Multi-Port Fuel Injection (SFI)

The fuel charging and controls system consists of the:

- Throttle Body (TB).
- fuel injectors.
- fuel rail.

The fuel charging and controls system is:

- a Sequential Multi-Port Fuel Injection (SFI) system.
- pulse-width modulated.
- Mass Air Flow (MAF) controlled.

Fuel is metered into each intake port in a sequential firing order. Fuel injectors pulse to follow engine firing order, in accordance with engine demand, on a tuned intake manifold.

The basic fuel requirement of the engine is determined from the data supplied to the PCM by the MAF sensor, which measures the amount of air being drawn into the engine.

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The various sensors detect any changes in the operating conditions and send signals to the PCM. This permits the PCM to control the opening duration (pulse width) of the fuel injectors and maintain optimum exhaust emission control and engine performance for all operating conditions.

Throttle Body (TB)

The TB:

- controls air supply to the intake manifold by electronically positioning the throttle plate.
- is not adjustable.

Fuel Injectors

The fuel injectors:

- are electronically operated by the PCM.
- atomize the fuel as the fuel is delivered.
- each have an internal solenoid that opens a needle valve, which injects fuel into the intake port in the cylinder head.
- are deposit resistant.

Fuel Rail

The fuel rail:

- receives fuel from the fuel supply tube.
- delivers fuel to the fuel injectors.

DIAGNOSTIC TESTS

FUEL CHARGING AND CONTROLS

Refer to the [Introduction - Gasoline Engines](#) article.

REMOVAL AND INSTALLATION

THROTTLE BODY

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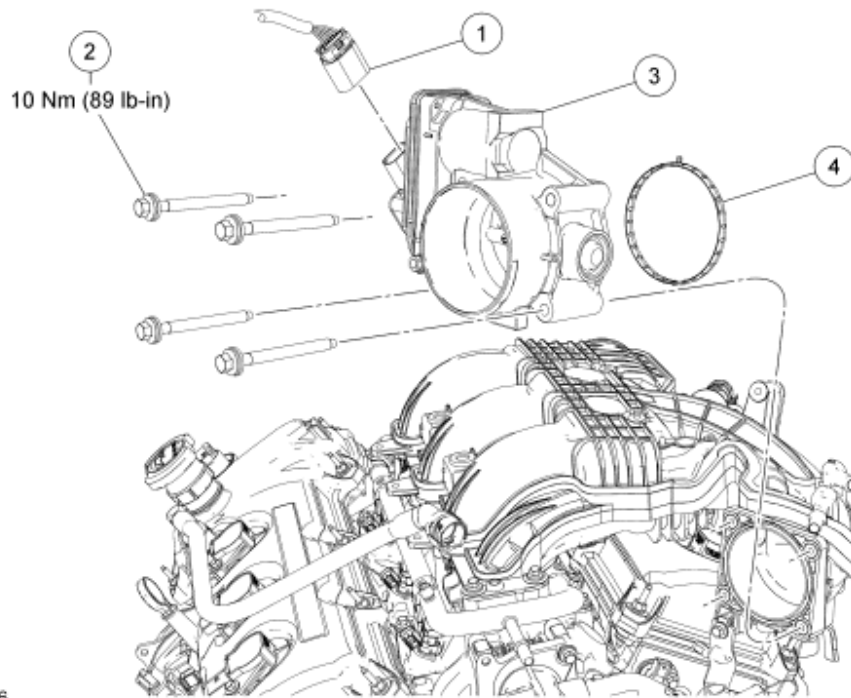


Fig. 1: Exploded View Of Throttle Body With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	Electronic throttle control electrical connector (part of 12B637)
2	W503285	Throttle body (TB) bolt (4 required)
3	9E926	TB
4	9E936	TB gasket

REMOVAL AND INSTALLATION

CAUTION: The throttle body (TB) must be removed from the vehicle to be cleaned. Do not hold the throttle plate open with any object that could scratch the bore or plate while servicing or cleaning the TB.

1. Remove the air cleaner outlet pipe. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
2. Disconnect the electronic throttle control electrical connector.
3. Remove the 4 bolts and the throttle body (TB).
 - Discard the TB gasket.
 - To install, tighten to 10 Nm (89 lb-in).

NOTE: Install a new TB gasket.

4. To install, reverse the removal procedure.

FUEL RAIL AND FUEL INJECTOR - EXPLODED VIEW

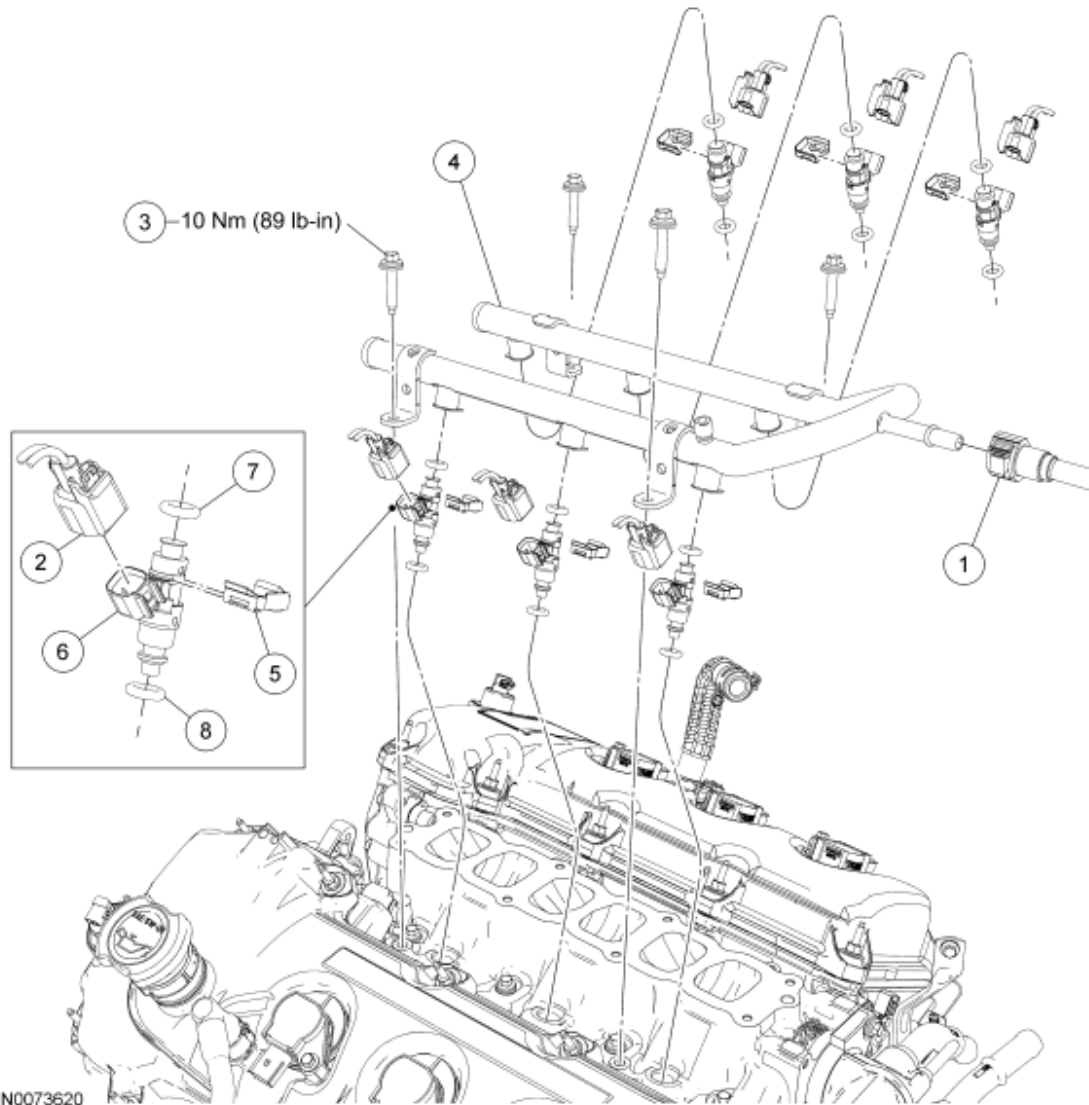


Fig. 2: Exploded View Of Fuel Rail & Fuel Injector With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	9J279	Fuel tube-to-fuel rail quick connect coupling
2	14A464	Fuel injector electrical connector (6 required) (part of 12B637)
3	W503279	Fuel rail bolt (4 required)
4	9F792	Fuel rail

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5	9N976	Fuel injector clip (6 required)
6	9F593	Fuel injector (6 required)
7	9229	Upper fuel injector O-ring seal (6 required)
8	9229	Lower fuel injector O-ring seal (6 required)

1. For additional information, refer to the procedures in this article.

FUEL RAIL

Material

Item	Specification
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (in Canada Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12) or equivalent	WSS-M2C930-A

REMOVAL

- WARNING:** Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.
- WARNING:** Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

1. Disconnect the battery ground cable. For additional information refer to **BATTERY, MOUNTING AND CABLES** article.
2. Release the fuel system pressure. For additional information, refer to **FUEL SYSTEM - GENERAL INFORMATION** article.
3. Remove the upper intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
4. Disconnect the fuel tube-to-fuel rail quick connect coupling. For additional information, refer to **FUEL SYSTEM - GENERAL INFORMATION** article.
5. Disconnect the 6 fuel injector electrical connectors.
6. Remove the 4 fuel rail bolts.
7. Remove the fuel rail and injectors as an assembly.
8. Remove the 6 fuel injector clips and the 6 fuel injectors.

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- Remove and discard the 12 fuel injector O-ring seals.

INSTALLATION

CAUTION: Use O-ring seals that are made of special fuel-resistant material. The use of ordinary O-rings seals can cause the fuel system to leak. Do not reuse the O-ring seals.

NOTE: The upper and lower O-ring seals are not interchangeable.

NOTE: Install new fuel injector O-ring seals and lubricate them with clean engine oil.

1. Install the 6 fuel injectors and the 6 fuel injector clips into the fuel rail.
2. Install the fuel rail and fuel injectors as an assembly.
3. Install the 4 fuel rail bolts.
 - Tighten to 10 Nm (89 lb-in).
4. Connect the 6 fuel injector electrical connectors.
5. Connect the fuel tube-to-fuel rail quick connect coupling. For additional information, refer to **FUEL SYSTEM - GENERAL INFORMATION** article.
6. Install the upper intake manifold. For additional information, refer to **ENGINE - 3.5L** article.
7. Connect the battery ground cable. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.